



Computer Science Department
COMP233-Discrete Mathematics / Spring 2026

Section #4 — Quiz#2 — Chapters 7,8.1

Time: 40 Minutes — Total Marks: 100 — weight: (10)

Student Name: _____ Student ID: _____

Question 1 (15 Marks): Fill **one** correct answer for each branch in the table below.

1. Let $f : \mathbb{Z} \rightarrow \mathbb{Z}$ be defined by $f(n) = 2n + 1$. Which statement is TRUE?			
a) f is onto but not one-to-one		b) f is one-to-one but not onto	
c) f is bijective		d) f has no inverse function	
2. The number of different functions from $A = \{1, 2, 3\}$ to $B = \{a, b\}$ is _____.			
a) $2^3 = 8$	b) $3^2 = 9$	c) $2 + 3 = 5$	d) $2 \times 3 = 6$
3. Let $f : A \rightarrow B$. Which statement is equivalent to saying that f is onto?			
a) For every $b \in B$, there exists a unique $a \in A$ such that $f(a) = b$		b) $\text{Range}(f) = B$	
c) If $f(a_1) = f(a_2)$, then $a_1 = a_2$		d) $ A \geq B $	
4. Let $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z} \times \mathbb{Z}$ be defined by $f(m, n) = (m + n, m - n)$. Then f is:			
a) one-to-one but not onto		b) onto but not one-to-one	
c) bijective		d) not a well-defined function	
5. Define $F : \mathcal{P}(\{1, 2, 3\}) \rightarrow \mathbb{Z}^{\text{nonneg}}$ by $F(S) = S$. The function F is:			
a) one-to-one and onto		b) one-to-one but not onto	
c) onto but not one-to-one		d) neither one-to-one nor onto	

Question 2 (50 Marks):

Consider the functions below.

- Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 4x - 1$. Determine whether f is a one-to-one correspondence. Prove your answer.
- If the inverse function f^{-1} exists, find $f^{-1}(x)$ and verify that $f(f^{-1}(x)) = x$.
- Let $g : \mathbb{Z} \rightarrow \mathbb{Z}$ be defined by $g(n) = n^2$. Disprove that g is one-to-one by giving an explicit counterexample.
- Consider the proposed function $w : \mathbb{Q} \rightarrow \mathbb{Q}$ defined by $w\left(\frac{m}{n}\right) = m + n$, where $m, n \in \mathbb{Z}$ and $n \neq 0$. Determine whether w is well-defined. If it is not, provide a counterexample and explain why the definition fails.

Question 3 (35 Marks): Let $A = \{2, 3, 4\}$ and $B = \{2, 6, 8\}$. Define the relation R from A to B by

$$x R y \iff x \mid y.$$

- List all ordered pairs of R .
- Find R^{-1} and list its ordered pairs.
- Draw the arrow diagram of R^{-1} .

Please, use the answer sheet form ONLY!
Good Luck!



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Answer Sheet

Question 1 (15 Marks):

Fill your final answers clearly in the table below.

Question	1	2	3	4	5
Answer					

Question 2 (50 Marks):

Q2.(a) *Answer:*

Q2.(b) *Answer:*

Q2.(c) *Answer:*

Q2.(d) *Answer:*

Question 3 (35 Marks):**Q3.(a)** *Answer:*

Q3.(b) *Answer:*

Q3.(c) *Answer:*

Extra Paper

Clearly label which question you are answering.